

Heterogeneous Verification for Multi-Step Decentralized Workflows

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Abstract

Financial and confidential automation relies on trusted intermediaries, imposing fees and custody risks that limit practical applications. Blockchains verify onchain operations but cannot verify automation requiring confidential execution, external data, or multi-step coordination. We propose a verification system where each operation uses the mechanism appropriate to its computation. The system combines cryptographic proofs for blockchain operations, hardware attestations for confidential execution, and stake-backed attestations for external interactions into workflow attestations. Validators broadcast attestations backed by stake, maintaining verifiability as long as honest validators control sufficient stake to make fraud economically irrational. This enables verifiable automation for financial operations, multi-party workflows, and confidential computation requiring trusted intermediaries.

1 Introduction

Execution of agreements and coordination of complex operations have come to rely almost exclusively on trusted intermediaries to handle financial transactions, legal processes, and confidential data. While these intermediaries work well enough for simple cases, they suffer from the inherent vulnerabilities of the trust-based model. Processes that handle value cannot avoid custody risk, since intermediaries must hold private keys and can be hacked, bankrupted, or compromised. The cost of custody manifests in billions of dollars lost to exchange failures and stolen assets, limiting the scale at which financial operations can be automated and cutting off entire classes of high-value coordination. Legal execution faces similar constraints—a will specifying asset distribution, guardianship, and property transfers requires coordinating executors, courts, property registries, and beneficiaries over months or years, with no mechanism for verifiable automation. Confidential processes compound these problems further, as healthcare workflows expose patient data to cloud providers, enterprise systems surrender control to vendors, and sensitive operations reveal private information to platforms. With these dependencies comes concentration of power. Organizations and individuals must accept vendor lock-in and platform rules, granting more access and control than they would otherwise. A certain percentage of custody losses, contested estates, data breaches, and censorship is accepted as unavoidable. These risks can be avoided through manual processes with in-person verification, but no mechanism exists for trustless execution at scale without centralized operators.

What is needed is a verification system based on cryptographic attestations instead of trust, allowing processes to execute across multiple steps and parties without centralized intermediaries. Workflows combining cryptographic proofs for blockchain operations, hardware attestations for confidential execution, and stake-backed attestations for external interactions would protect against custody risk while preserving confidentiality and maintaining verifiability across heterogeneous trust requirements. In this paper, we propose a heterogeneous verification system where validators attest

to each workflow step using mechanisms appropriate to the computation and stake economic value against their attestations. Fraudulent attestations can be challenged through onchain fraud proofs. The system is secure as long as honest validators control sufficient stake to make fraud economically irrational.

References

- [1] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.